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$$\begin{aligned} & \left(\operatorname{cis} \left(\frac{\pi}{3} \right) \right)^{100} \\ &= \operatorname{cis} \left(100 \left(\frac{\pi}{3} \right) \right) \\ &= \operatorname{cis} \left(16 \frac{6\pi}{3} + \frac{4\pi}{3} \right) \\ &= \operatorname{cis} \left(16 \frac{2\pi}{3} + \frac{4\pi}{3} \right) \\ &= \operatorname{cis} \left(\frac{4\pi}{3} \right) \\ &= \cos \left(\frac{4\pi}{3} \right) + i \sin \left(\frac{4\pi}{3} \right) \\ &= -\frac{1}{2} - \frac{\sqrt{3}}{2}i \end{aligned}$$

References